

COPD Definition and Diagnosis

Definition COPD is a heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production and/or exacerbations) due to abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction 1, 2.

Clinical Indicators for Diagnosis A diagnosis should be considered if any of the following are present:

- **Dyspnea:** Progressive over time, worse with exercise, and persistent 3.
- **Chronic cough:** May be intermittent and non-productive 3.
- **Sputum production:** Regular production for 3+ months in two consecutive years 4.
- **Recurrent lower respiratory tract infections** 3.
- **History of risk factors:** Tobacco smoke (including local preparations), smoke from home cooking/heating fuels, occupational dusts/vapors/fumes, or host factors (e.g., low birthweight, prematurity) 3.

Diagnostic Confirmation

- **Mandatory Test:** Forced spirometry is required to establish the diagnosis 5, 6.
- **Criterion:** Presence of a **post-bronchodilator FEV1/FVC < 0.7** confirms persistent airflow obstruction 5, 7, 8.
- **Refinement:** If the ratio is between 0.6 and 0.8, spirometry should be repeated on a separate occasion 9.

Source: GOLD Section 1 & 2; Pages 4, 21, 26, 29.

Spirometric Grades and Classification

Classification of Airflow Obstruction Severity Based on post-bronchodilator FEV1 (% predicted) in patients with FEV1/FVC < 0.7:

- **GOLD 1 (Mild):** FEV1 ≥ 80% predicted 10.
- **GOLD 2 (Moderate):** 50% ≤ FEV1 < 80% predicted 10.
- **GOLD 3 (Severe):** 30% ≤ FEV1 < 50% predicted 10.
- **GOLD 4 (Very Severe):** FEV1 < 30% predicted 10.

Source: GOLD Section 2; Page 37; Figure 2.10.

Initial Assessment and GOLD Grouping (ABE)

Symptom Assessment Tools

- **mMRC Scale:** Measures breathlessness (0 to 4) 11.
- **CAAT™/CAT™:** 8-item questionnaire measuring health status (0 to 40) 12.

GOLD ABE Classification (For Treatment-Naïve Patients) Patients are assigned to groups based on symptom burden and exacerbation history in the previous year:

- **Group A:** 0 or 1 moderate exacerbation (not requiring hospital admission) AND mMRC 0–1 or CAT < 10 13.
- **Group B:** 0 or 1 moderate exacerbation (not requiring hospital admission) AND mMRC ≥ 2 or CAT ≥ 10 13.

- **Group E:** ≥ 2 moderate exacerbations OR ≥ 1 exacerbation leading to hospital admission (regardless of symptom scores) 13, 14.

Biomarker Thresholds

- **Blood Eosinophil Count:** Recommended to guide Inhaled Corticosteroid (ICS) use 15, 16.
- **Thresholds:** Counts < 100 cells/ μ L suggest ICS is unlikely to be effective; counts ≥ 300 cells/ μ L identify patients with the greatest likelihood of ICS benefit 17, 18.

Source: GOLD Section 2; Pages 40-43; Figure 2.13.

Initial Pharmacological Treatment

Preferred Initial Therapies

- **Group A:** A bronchodilator (short or long-acting) 19, 20.
- **Group B:** Dual bronchodilation with **LABA + LAMA** 19, 20.
- **Group E:** Dual bronchodilation with **LABA + LAMA**.
- **Triple Therapy (LABA + LAMA + ICS):** Consider as initial therapy if blood eosinophils are ≥ 300 cells/ μ L 19, 21.

Source: GOLD Section 3; Pages 64-65; Figure 3.8.

Follow-up Pharmacological Management and Escalation

Decision Pathways Treatment adjustments are based on the predominant treatable trait:

Persistent Dyspnea or Exacerbations 16, 22.

1. Persistent Dyspnea Pathway

- If on monotherapy (LABA or LAMA) -> escalate to **LABA + LAMA** 22.
- If symptoms persist on LABA + LAMA -> consider switching device/molecules or adding ensifentrine (where available) 23.

2. Exacerbation Pathway

- If on monotherapy (LABA or LAMA) -> escalate to **LABA + LAMA** 24.
- If on LABA + LAMA:
- Escalate to **Triple Therapy (LABA + LAMA + ICS)** if blood eosinophils are ≥ 100 cells/ μ L 24.
- If eosinophils < 100 cells/ μ L, consider adding **Roflumilast** (if FEV1 $< 50\%$ and chronic bronchitis) or **Azithromycin** (preferentially in former smokers) 24, 25.
- If on LABA + LAMA + ICS:
- Consider adding **Dupilumab** (if chronic bronchitis and eos ≥ 300 cells/ μ L) or **Mepolizumab** (if eos ≥ 300 cells/ μ L) 26-28.

Source: GOLD Section 3; Pages 66-68; Figure 3.9.

Non-Pharmacological Management and Prevention

Smoking Cessation

- **Recommendation:** Actively pursue for all patients (Evidence A). Combination of counseling and pharmacotherapy is most effective 29, 30.
- **Pharmacotherapy:** Nicotine replacement, varenicline, bupropion, or nortriptyline 31, 32.

Vaccinations (Evidence B)

- **Influenza:** Annual routine vaccination 33.
- **COVID-19:** According to national recommendations 34.
- **Pneumococcal:** One dose of PCV20 or PCV21 is recommended 35, 36.
- **RSV:** Recommended for patients ≥ 60 years or with chronic lung disease 37.
- **Other:** Tdap (pertussis/tetanus/diphtheria) for those not vaccinated in adolescence; routine Zoster (shingles) vaccine 34, 38.

Pulmonary Rehabilitation

- **Indication:** Patients with high symptom burden or risk of exacerbations (Groups B and E) (Evidence A) 39, 40.
- **Duration:** Optimum benefits achieved from programs lasting **6 to 8 weeks** 41.

Source: GOLD Section 3; Pages 53, 56-58, 74.

Oxygen Therapy and Ventilatory Support

Long-Term Oxygen Therapy (LTOT) Indicated for stable patients with severe resting chronic hypoxemia, defined as:

- **PaO₂ $\leq$ 55 mmHg (7.3 kPa) or SaO₂ $\leq$ 88%** (confirmed twice over 3 weeks) 42.
- **PaO₂ 55–60 mmHg (7.3–8.0 kPa)** if there is evidence of pulmonary hypertension, peripheral edema (congestive heart failure), or polycythemia (hematocrit $> 55\%$) 42.
- **Goal:** Titrate to keep **SaO₂ $\geq$ 90%** 43.

Ventilatory Support

- **CPAP:** Indicated for patients with comorbid COPD and Obstructive Sleep Apnea (Evidence B) 44.
- **NPPV:** May be considered in patients with persistent hypercapnia (**PaCO₂ $>$ 53 mmHg**) 2–4 weeks after hospital discharge for an acute exacerbation 45.

Source: GOLD Section 3; Pages 76, 78.

Management of Acute Exacerbations

Definition and Severity (Rome Proposal) An acute event characterized by worsened symptoms (dyspnea, cough, sputum) over ≤ 14 days 46.

- **Mild:** 0-10 VAS < 5 ; RR < 24 ; HR < 95 ; SaO₂ $\geq 92\%$ (change $\leq 3\%$) 47.
- **Moderate:** At least 3 of 5: VAS ≥ 5 ; RR ≥ 24 ; HR ≥ 95 ; SaO₂ $< 92\%$ (change $> 3\%$); CRP ≥ 10 mg/L 47.
- **Severe:** Symptoms of moderate plus ABG showing hypoxemia (**PaO₂ $\leq$ 60 mmHg**) and/or hypercapnia/acidosis (**PaCO₂ $>$ 45 mmHg, pH $<$ 7.35**) 48.

Pharmacological Treatment

- **Bronchodilators:** Inhaled SABAs, with or without short-acting anticholinergics (Evidence C) 49, 50.
- **Systemic Corticosteroids:** **40 mg prednisone equivalent per day for 5 days** (Evidence A) 49, 51.
- **Antibiotics:** Recommended for **5 days** (Evidence B) if patients have purulent sputum, prior positive sputum culture, or require mechanical ventilation 49, 52, 53.

- **Contraindication:** Methylxanthines (theophylline/aminophylline) are not recommended during exacerbations due to side effects 54, 55.

Respiratory Support

- **High-Flow Nasal Therapy (HFNT):** First mode for acute hypoxemic failure 56.
- **NIV:** Initial mode for acute respiratory failure (Evidence A). **Indications:** Acidosis ($\text{pH} \leq 7.35$ and $\text{PaCO}_2 \geq 45$ mmHg), severe dyspnea with muscle fatigue, or persistent hypoxemia despite oxygen 57, 58.

Source: GOLD Section 4; Pages 92, 98-99, 103-107.